



Canada and Mexico, are raising costs and disrupting supply chains. The withdrawal of certain state-level incentives has further softened demand.

By contrast, China's domestic manufacturers are expanding their dominance, while Western OEMs now hold a smaller share of the market. India is strengthening its position through Production-Linked Incentive (PLI) schemes, including a \$3.1 billion programme for automobiles and components and a ₹180 billion initiative to build 50GWh advanced battery capacity, reinforcing local manufacturing and supply chains.

Beyond infrastructure and policy, product trends are evolving as well. Battery-electric and hybrid vehicles are leading market growth, with consumers increasingly choosing mid-size models over very small or very large vehicles. This shift reflects practical mobility and logistics needs, particularly in North America and India. Together, these factors indicate that while regional growth rates may vary, the global EV transition is firmly underway.

Key technology and infrastructure trends shaping EV adoption

Charging infrastructure remains the single most important factor in accelerating EV adoption over the next five years. In North America, networks are expanding but still face bottlenecks. The International Energy Agency estimates that the United States had around 204,000 public chargers by 2024. However, progress under the \$7.5 billion federal charging programme has

Challenges in launching a new EV model

- **Talent shortage.** Companies struggle to find engineers with hands-on experience in emerging EV systems
- **Supplier onboarding.** Tight timelines require NDAs, regulatory approvals, system integration, and engineering alignment, often delaying production
- **Coordination bottlenecks.** Many Indian suppliers are willing to participate, but slow collaboration slows integration timelines
- **Cost pressures.** Tariffs and high interest rates, especially in the US, increase production costs and affect adoption
- **Strategic outsourcing.** Sourcing select components from cost-competitive markets like India can reduce operational pressure and support wider EV adoption

been slow, with only a few hundred charging ports delivered by mid-2025. The rollout of the North American Charging Standard is helping improve interoperability between fast chargers, especially for DC Level 3 charging, which is critical for long-distance travel. Governments are also focusing on grid integration through software platforms and energy management systems to ensure charging demand does not strain power networks.

In India, growth has been rapid. The public charging network expanded from just a few thousand stations in 2022 to tens of thousands by late 2025, reflecting strong compound annual growth. Even so, infrastructure remains a constraint. Estimates suggest that India may require around 1.3 million charging stations by 2030 to meet projected demand. Fast DC charging, battery-swapping pilots for two-wheelers and three-wheelers, and solar-powered charging stations connected to smart grids are emerging as important trends.

Battery innovation will also shape adoption. Solid-state batteries promise longer driving range and lower maintenance, reducing

operating costs and making EVs more practical as fuel prices rise, although replacement complexity remains a key limitation. Vehicle-to-grid systems add value by allowing EVs to feed electricity back into the grid, supporting energy balancing and improving grid stability.

On the cost front, lithium-iron phosphate batteries dominate the mass market due to their balance of cost and performance. Sodium-ion batteries are emerging as a lower-cost alternative, offering faster charging and shorter refuelling times. Solid-state batteries are expected to remain focused on premium, high-density applications, while sodium-ion technology could help reduce costs for mass-market vehicles.

Indian EV manufacturers face distinct challenges compared to global players. Infrastructure gaps continue to affect consumer confidence, especially for long-distance travel. Surveys indicate that many buyers still perceive EVs as expensive, with charging time and limited station availability adding to hesitation, thereby slowing adoption relative to mature global markets.

Supply chain and technology gaps are another concern. Advanced battery engineering is still in its early stages in India, with more than 70% of lithium-ion cell requirements imported mainly from China, South Korea, and Japan. Global manufacturers such as Tesla, Lucid, and BYD already offer vehicles with ranges approaching 700 kilometres per charge, setting performance

Embedding sustainability without slowing EV launches

- **Integrate early.** Identify regulatory and homologation dependencies at the start
- **Cross-functional collaboration.** Align engineering, operations, and compliance teams
- **Proactive planning.** Anticipate risks and plan around performance, quality, cost, and timelines
- **Data-driven risk registers.** Track potential issues to avoid late-stage surprises
- **Sustainable delivery.** Balance environmental goals with on-time product launches